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ARTICLE

Design of a novel continuous flow reactor for low pH viral inactivation

Stephanie A. Parker¹  | Linus Amarikwa^{1,2} | Kevin Vehar¹ | Raquel Orozco² | Scott Godfrey² | Jon Coffman³ | Parviz Shamlou¹ | Cameron L. Bardliving¹

¹ Keck Graduate Institute, Amgen Bioprocessing Center, Claremont, California

² Boehringer Ingelheim, Department of Process Science, Fremont, California

³ Boehringer Ingelheim, Department of Global Innovation and Technology, Fremont, California

Correspondence

Stephanie A. Parker, Keck Graduate Institute, Amgen Bioprocessing Center, Claremont, CA 91711.

Email: stephanie_parker@kgi.edu

Abstract

Insufficient mixing in laminar flow reactors due to diffusion-dominated flow limits their use in applications where narrow residence time distribution (RTD) is required. The aim of this study was to design and characterize a laminar flow (Re 187.7–375.5) tubular reactor for low pH viral inactivation with enhanced radial mixing via the incorporation of curvature and flow inversions. Toward this aim, the reactor described here, Jig in a Box (JIB), was designed with a flow path consisting of alternating 270° turns. The design was optimized by considering the strength of secondary flows characterized by the Dean No., the corresponding secondary flow development length, and the reactor turn lengths. Comprehensive CFD analysis of the reactor centerline velocity profile, cross-sectional velocity, and secondary flow streamlines confirmed enhanced radial mixing due to secondary flows and changes in flow direction. For initial CFD and experimental studies the reactor was limited to a 16.43 m length. Pulse tracer studies for the reactor were computationally simulated and experimentally generated to determine the RTD, RTD variance, and minimum residence time for the tracer fluid elements leaving the reactor, as well as to validate the computational model. The reactor was scaled length wise to increase incubation time and it was observed that as the reactor length increases the RTD variance increases linearly and the dimensionless RTD profile becomes more symmetrical and tighter about the mean residence time.

KEYWORDS

axial dispersion, computational fluid dynamics, continuous processing, low pH viral inactivation, residence time distribution

1 | INTRODUCTION

1.1 | Continuous bioprocessing

In response to the increased demand for biologics (Aggarwal, 2014) and competition from biosimilar products, the biopharmaceutical

industry has begun moving from batch to continuous manufacturing (Gronemeyer, Ditz, & Strube, 2014; Hausner, Lustri, & Brodbeck, 2016; Klutz, Magnus et al., 2015; Mothes, Pezzini, Schroeder-Tittmann, & Villain, 2016; Schaber et al., 2011; Subramanian, 2015; Warikoo et al., 2012; Zydney, 2016). Across industries, the advantages of continuous over batch are consistent and include steady state operation, process flexibility, reduction in the ratio of equipment size to production capacity, streamlined process control, improved product quality control, and reduced capital costs (Anderson, 2001; Konstantinov &

Abbreviations: CFD, computational fluid dynamics; CVI, continuous viral inactivation; JIB, jig in a box; MRT, minimum residence time; ROC, radius of curvature; RTD, residence time distribution; VI, viral inactivation.

Cooney, 2015; Reay, Ramshaw, & Harvey, 2013; Stankiewicz & Moulijn, 2000; Tanner, 1998). Advancements have been made to the mammalian cell production platform in the form of perfusion bioreactors that allow for a continuous stream of product to be removed for downstream processing (Ahn, Jeon, Jeong, Lee, & Yoon, 2008; Bonham-Carter & Shevitz, 2011; Gottschalk, Brorson, & Shukla, 2012; Gray, Chen, Howarth, Inlow, & Maiorella, 1996; Voisard, Meuwly, Ruffieux, Baer, & Kadouri, 2003). Additionally, methods for continuous multi-column chromatography are currently being explored for use in continuous downstream purification and polishing steps (Bisschops, Frick, Fulton, & Ransohoff, 2009; Girard et al., 2015; Steinebach, Muller-Spath, & Morbidelli, 2016). Low pH based viral inactivation (VI) is typically placed between the Protein A capture and polishing chromatography steps (Liu, Ma, Winter, & Bayer, 2010; Shukla, Hubbard, Tressel, Guhan, & Low, 2007). During this step, the product is pooled and held in a tank. This static process presents a major bottle neck for end-to-end continuous downstream processing that has not currently been remedied.

1.2 | Continuous viral inactivation

Viral safety is mandated for protein therapeutics produced in mammalian cells and viral clearance procedures are highly regulated (FDA, 1998). Low pH VI is a highly effective and common method used in monoclonal antibody purification processes, consistently clearing greater than 4 log(10) of large enveloped viruses, including endogenous retroviruses (Brorson et al., 2003; Shukla et al., 2007). The 2012 ASTM Standard Practice for Process for Inactivation of Retrovirus by pH prescribes the following conditions for low pH viral inactivation: pH ≤ 3.6 , at $\geq 15^{\circ}\text{C}$, for ≥ 30 min in a system specific buffer at the specified pH, providing ≥ 5 log reduction value (LRV).

Two methods have been proposed to transition from batch to continuous low pH VI: i) a semi-continuous repetitive batch where the product is continuously pooled in tanks and ii) a continuous tubular reactor (Klutz, Kurt, Lobedann, & Kockmann, 2015; Zydney, 2016). For this work, we designed a continuous viral inactivation (CVI) tubular reactor with in-line static mixers to adjust the pH of the product as it flows into and out of the reactor, Figure 1. The static mixers will be

used to first lower the pH of the protein A elute to the desired viral inactivation pH, by mixing with a low pH buffer, and subsequently to increase the pH after inactivation for further processing steps, as shown in Figure 1.

1.3 | Continuous viral inactivation (CVI) reactor design considerations

To minimize shear forces on the product and pressure drop across the reactor, the flow velocity is limited to the laminar regime. Limiting CVI pressure drop will also reduce the likelihood that back pressure from the reactor will impact the flow of product coming from the Protein A column or flow to further downstream process chromatography steps (Schoenmakers & Uunk, 1987). The major challenge of operating at low Reynolds (Re) number (Equation 1) is axial dispersion, as laminar flow is characterized by a parabolic velocity profile, where fluid elements at the center of the tube travel faster than the elements near the wall resulting in a broad Residence Time Distribution (RTD). Thus, axial dispersion in the CVI reactor must be characterized to ensure sufficient incubation time. Further, incorporating curvature into the flow path produces secondary flows that enhance radial mixing and reduce axial dispersion (Janssen, 1976; Pegoraro, Marangoni, & Gut, 2012; Saxena & Nigam, 1983; Vashisth, Kumar, & Nigam, 2008). Additionally, passive mixing induced via flow inversions or shifts in peak velocity, can be incorporated to reduce axial dispersion (Klutz, Kurt et al., 2015; Kurt, Gelhausen, & Kockmann, 2015; Levenspiel, 1999). Here, we operated at low Reynolds numbers to negate pressure concerns and incorporated curvature and flow inversions along the flow path to minimize axial dispersion in our CVI reactor.

1.4 | Computational fluid dynamics

Computational Fluid Dynamics (CFD) modeling was employed by simulating pulse tracer experiments and the steady state flow field velocity profiles were generated to understand the complex flow behavior of our CVI reactor (Francis & Haynes, 2009; Kelly, 2008; Lode, Rosenfeld, Yuan, Root, & Lightfoot, 1998; Manninen, Gorshkova, Immonen, & Ni, 2013). Extensive pulse tracer simulations were

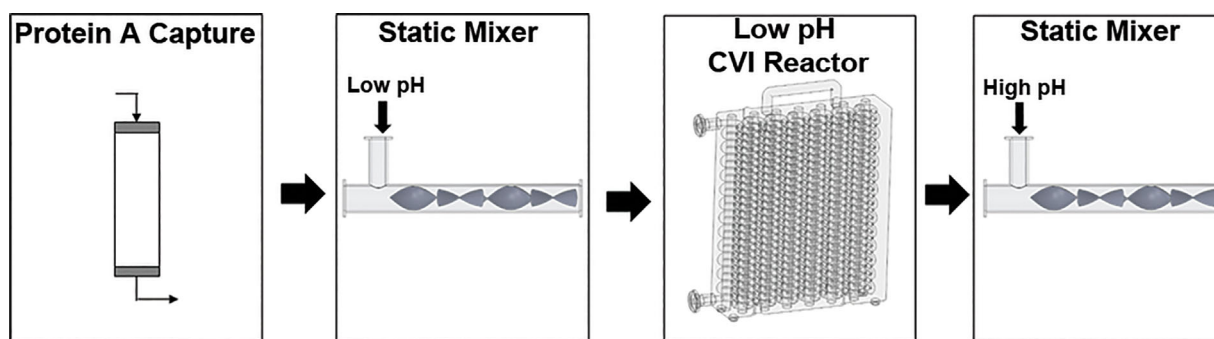


FIGURE 1 Proposed continuous low pH viral inactivation scheme. Continuous flow from protein A capture elution will pass through a static mixer to adjust pH, followed by an in-line tubular Continuous Viral Inactivation (CVI) reactor (JIG in a Box [JIB]), and finally another static mixer for pH adjustment before moving to further downstream processing steps

conducted to predict and characterize the RTD for the CVI reactor at varying flow rates. Simulated RTDs were used to determine the reactor path length that will yield a specified incubation time and were validated experimentally (Thakre, Phanikumar, Khare, & Joshi, 1999). The steady state flow field CFD results were used to determine the impact of the flow path curvature and flow inversions on radial mixing.

2 | THEORY

2.1 | Design physics: Dean vortices

Laminar flow occurs at Reynolds Numbers (Re) less than 2000. The Re for flow in a pipe is defined by Equation 1 below, where ρ is density, v is average velocity, d is the tube diameter, and μ is kinematic viscosity.

$$Re = \frac{\rho v d}{\mu} \quad (1)$$

For steady motion of an incompressible fluid in a curved pipe the strength of the secondary flows are characterized by a non-dimensional parameter, the Dean No (D), described by Equation (2) where d is the tube inner diameter and R is radius of curvature of the flow path (Dean, 1927, 1928).

$$D = Re \sqrt{\frac{d}{2R}} \quad (2)$$

2.2 | Residence time distribution (RTD)

The RTD of the reactor can be represented such that the area under the curve is unity, E curve, and is defined by Equation (3), where C is concentration of tracer at the outlet and t is time (Levenspiel, 1999).

$$\int_0^{\infty} E(t) dt = \frac{C(t)}{\int_0^{\infty} C(t) dt} = 1 \quad (3)$$

Axial dispersion is characterized by the mean residence time, $\bar{t}$, and variance, σ^2 , defined by Equations (4 and 5), respectfully.

$$\bar{t} = \frac{\int_0^{\infty} t C dt}{\int_0^{\infty} C dt} \quad (4)$$

$$\sigma^2 = \frac{\int_0^{\infty} t^2 C dt}{\int_0^{\infty} C dt} - \bar{t}^2 \quad (5)$$

The E curve can be expressed in dimensionless form, $E(\theta)$, as described in Equation (6), where θ is dimensionless time and $\theta = t/\bar{t}$. Variance can be expressed in dimensionless time by dividing the variance by the squared value of the mean residence time, $\sigma_{\theta}^2 = \sigma_t^2/\bar{t}^2$. Additionally, the step output, F curve, is found by integrating the

corresponding E curve, as described in Equation (7).

$$E_{\theta} = \bar{t} E \quad (6)$$

$$F = \int_0^{\theta} E_{\theta} d\theta = \int_0^t E(t) dt \quad (7)$$

Using the definition proposed by Kurt et al. (2015), the relative width (R_w) or narrowness of the RTD peak can be characterized by using the dimensionless $F(\theta)$ step output to obtain the dimensionless times corresponding to the dimensionless concentrations 0.005 (0.05%) and 0.995 (99.5%), and the R_w is equal to the ratio of the values, as described in Equation (8).

$$R_w = \frac{\theta_{0.5\%}}{\theta_{99.5\%}} \quad (8)$$

3 | REACTOR DESIGN

Previously, we determined that flow around turns in combination with shifting the direction of curvature from one turn to the next, enhances radial mixing and reduces axial dispersion in a tubular reactor operated at low Re (Orozco et al., 2017). Based on this observation our CVI reactor, the Jig in a Box (JIB), consists of alternating 270° turns. The JIB was designed for use with pilot scale equipment tubing and thus we selected an inner tube diameter of 0.635 cm (Orozco et al., 2017).

To select the radius of curvature (ROC) for our JIB design we numerically investigated the relationship between the turn length and the strength of the secondary flows. We used the mathematical correlation developed by Springer, Carretier, Veyret, and Moulin (2009) for the length required for the secondary flow in a torus, at low Re (0–400), to become fully developed (L_{DT}), as described in formula (9), where d_i is the internal diameter and d_c is the coil diameter in meters.

$$L_{DT} = 0.322 \times d_c^{0.31} \times Re^{0.59} \times d_i^{0.76} \quad (9)$$

To determine the optimal turn radius for the JIB, the Dean No and the ratio of the turn length to L_{DT} value at varying d_c values were computed with respect to a range of radii at 100 ml/min (0.0526 m/s), the top of our operating range, as this flow rate will result in the largest L_{DT} value (Figure 2).

The design criteria of alternating 270° turns restricts the ROC to approximately 0.85 cm when allowing for a tube wall thickness of 0.5 mm. This ROC value results in an arc length of 4.0 cm which is lower than the predicted L_{DT} value of 6.43 cm, as shown in Figure 2. However, incipient secondary motions in the boundary layer cause reversal of the initial distribution of the shear-rate after a distance of approximately one diameter (Caro, 1979) and from there on the maximum shear rate occurs at the outside wall, as when the flow is fully developed. As seen in Figure 2, the ROC value of 1 cm, exhibits relatively high Deans No (>210) at

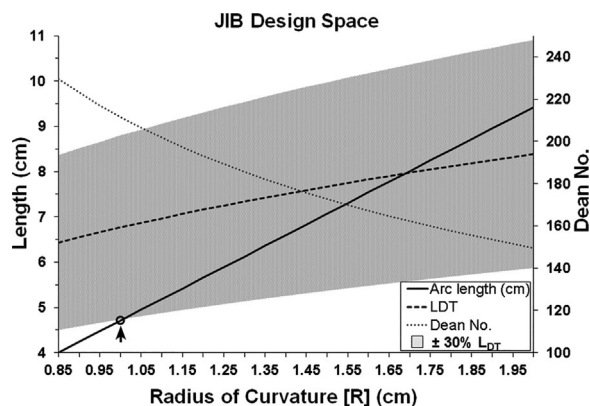


FIGURE 2 JIB Radius of Curvature vs. arc length of the JIB alternating 270° turns, requisite secondary flow development turn length (LDT), and characteristic Dean No at a flow rate of 100 ml/min (0.0526 m/s) and tube inner diameter of 0.635 cm. The arc length (Length) was calculated for a 270° turn with radii of curvature (ROC) ranging from 0.85–2.0 cm. The estimated length required for secondary flows to become fully developed (LDT) in a torus at the given ROC values was calculated using the mathematical correlation developed by Springer et. al (Equation 9). The shaded area represents a range of within 30% of the LDT value. The ROC selected for this study, 1.0 cm, is indicated by the black arrow. As shown in the figure, a 1.0 cm ROC corresponds to the highest Dean No. with a turn length of 4.71 cm which is within approximately 30% (30.3%) of the turn length required for the flow to become fully developed, 6.76 cm

0.0526 m/s, and results in an arc length of 4.71 cm which is within 30.3% of the L_{DT} of 6.76 cm, and a ratio of turn length to $L_{DT} > 0.6$, and was selected for the JIB design.

4 | MATERIALS AND METHODS

4.1 | Construction and design of CVI reactor

SOLIDWORKS (SolidWorks Corporation, Waltham, MA) Computer-aided design (CAD) software was used to generate the CVI reactor design. The design was subsequently 3D printed from the CAD file by ProX 800 SLA Technology (ProX 800 SLA Technology, 3D Systems, Rock Hill South, CA).

The inner diameter of the JIB is 0.635 cm and the design was evaluated for an operational volumetric flow rate (Q) range of 50–100 ml/min (0.0263–0.0526 m/s and Re 187.7–375.5). The design consists of 325 alternating 270° turns with a 1 cm ROC, resulting in a serpentine pattern with turn lengths of 4.71 cm, as shown in Figure 3. The vertical (L1) and horizontal (L2) center to center distance between the 270° turns are 1.5 and 1.375 cm, respectfully, Figure 3a. To accommodate a compact design the turns are segmented vertically into 26 stacked layers with a tube center to center distance of 0.255 cm. Each layer consists of 12.5 turns each and are connected by 180° turns, as seen in Figure 3c. The 3D printed JIB footprint is 20 × 4.9 × 23 cm and contains a flow path length of approximately 16.43 m resulting in a 520 ml flow volume.

To allow for alterations to the path length and incubation time, flanged connectors were designed to connect the JIBs in series, as shown in Figure 3d.

4.2 | Tracer experiments

Pulse tracer experiments with the CVI reactor consisted of first flushing the JIB with Milli-Q water, (Barnstead Nanopure Water Purification System, Thermo Scientific, Waltham, MA), followed by a 13 ml pulse injection of 50 mg/ml Riboflavin, and finally chasing with Milli-Q water via a peristaltic pump (520 IP31, Watson-Marlow Flow Technology Group, Lake Forest, CA). The absorbance of the tracer at the outlet was measured using a FlowVPE (C Technologies, Inc. New Jersey, NY) flow cell unit integrated with a Cary 60 UV-Vis Spectrophotometer (Agilent Technologies, Santa, Clara, CA) at a wavelength of 372 nm. Silicone tubing 0.25" ID (Masterflex 96410-24, Cole-Parmer, Vernon Hills, IL) was used to connect the pump to the reactor and reactor to the Spectrophotometer.

4.3 | Computational fluid dynamics

ANSYS Fluent Workbench v. 17.2 species transport model was used to simulate JIB tracer studies. For this work, the SOLIDWORKS CAD geometry of the CVI reactor was directly imported into ANSYS. The CFD geometry was matched with the experimental setup by adding a straight volume of tubing at the entrance and before the measurement point. The 3D flow region was divided into hexahedral dominate finite volumes with higher resolution mesh elements employed at the tube walls, to capture near wall flow physics, using ANSYS® Meshing™. Specifically, an unstructured grid with inflation along the edge was applied to the outlet face and swept through the rest of the body, using an all quad free face mesh defined by the sweep number of divisions. Investigation of grid independence was performed using a global mesh refinement algorithm based on a 2.5 ratio of maximum change in cell volume, with a range of 1.5–6.5 million elements. Steady-state simulations monitoring pressure drop and outlet velocity were run at each level of grid refinement. Results of mesh independence studies demonstrated that 2.5 million elements was sufficient for further studies.

The computational model was first solved for steady state with convergence criteria of $1e^{-8}$ residuals for the velocity field. Next, the steady state velocity field was used to initialize a transient model where a 13 ml volume pulse tracer with mass diffusivity of $0.8 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ is injected into the reactor. To match the experimental setup the tracer is contained in 13 ml volume of straight tubing at the start of the simulation, or $t = 0$, the model then simulates the pulse tracer volume traveling through the reactor. A surface monitor at the outlet of the reactor recorded the mass-weighted molar concentration of the tracer at the outlet at each time step. ANSYS Fluent predicts the local mass fraction of each species, Y_i , through the solution of a convection-diffusion equation for the i th species using the

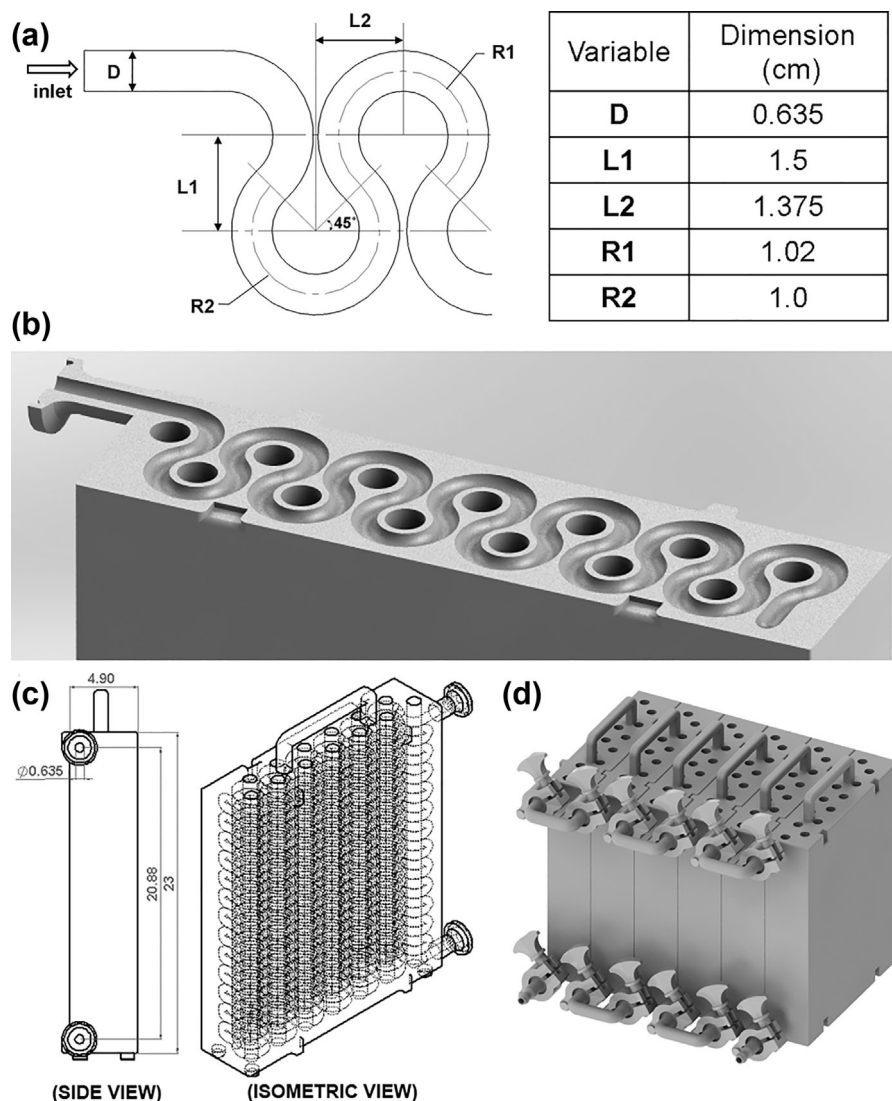


FIGURE 3 Continuous Tubular Reactor Design called Jig in a Box (JIB). a) Tubular flow path with dimensions. b) Cross Sectional view of the JIB flow path. c) Left: Side of chamber with dimensions (4.9 × 23 cm W × H) showing the entrance and exit of chamber and right: wireframe isometric view showing internal flow path consisting of 325 (270°) turns divided into 26 sections of 12.5 turns that are stacked vertically and connected by 25 (180°) bends. d) Six JIBs connected in series

Convection-diffusion Equation (10).

$$\frac{\partial}{\partial t}(\rho Y_i) + \nabla \cdot (\rho \mathbf{v} Y_i) = -\nabla \cdot \mathbf{J}_i \quad (10)$$

In Equation (10) J_i is the diffusion flux of species i due to gradients of concentration and $\mathbf{v}$, is the steady state velocity field. To approximate diffusion due to concentration gradients for mass diffusion in laminar flows ANSYS Fluent solves the Fick's Law dilute approximation written as:

$$\mathbf{J}_i = -\rho D_{i,m} \nabla Y_i \quad (11)$$

$D_{i,m}$ is the mass diffusion coefficient for species i in the mixture.

5 | RESULTS AND DISCUSSION

5.1 | Flow dynamics

The steady-state flow field of the JIB was computed using ANSYS solver and the flow dynamics of the system were characterized via the center plane velocity profile, cross sectional velocity profiles, and secondary flow patterns. Characterization of the JIB flow dynamics was used to determine the extent to which the addition of curvature in the flow path disrupts the uniform and highly ordered motion characteristic of the laminar flow regime. The JIB centerline velocity profile was measured at the center plane of the flow path perpendicular to the inlet, as shown in Figure 4a. Figure 4a shows the center plane velocity contours for the JIB at a volumetric flow rate

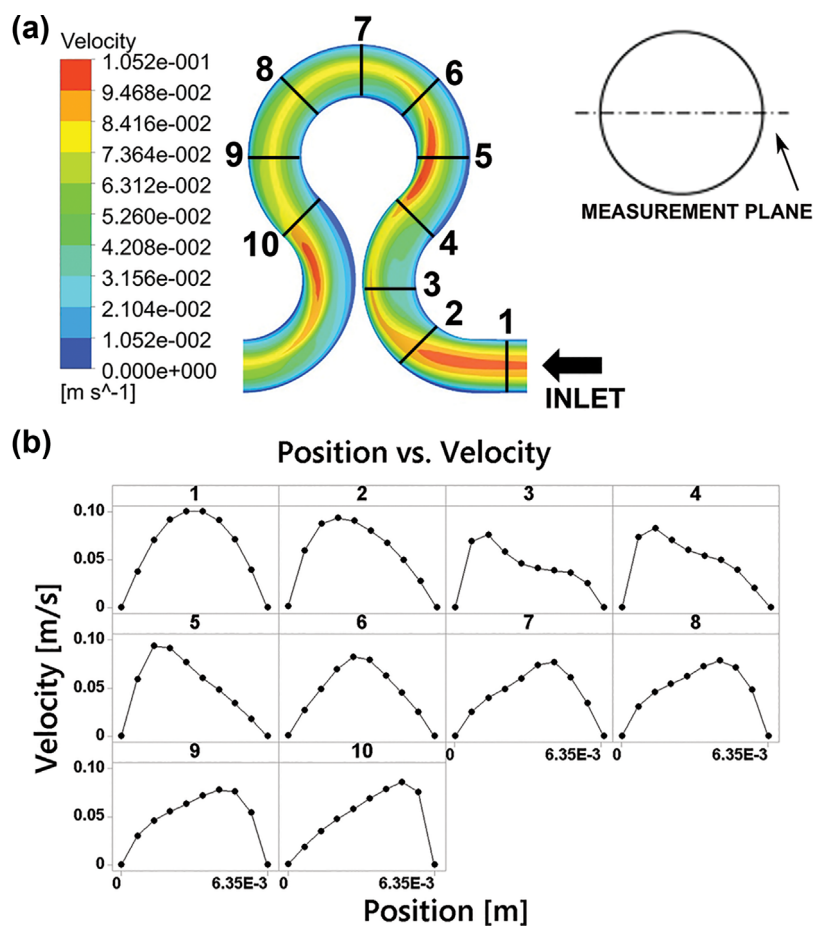


FIGURE 4 Computational fluid dynamics results for the center plane velocity contour and profiles for the Continuous Viral Inactivation (CVI) reactor or JIB. The velocity profile measured at the center plane, perpendicular to the flow direction, of the alternating 270° turns of the JIB flow path, where the inlet velocity is 0.0526 m/s (100 ml/min and $\text{Re } 375.5$). a) CFD velocity magnitude distribution contours at the center plane of the channel. The range of velocity magnitude is from 0 to 1.052 m/s , where the maximum velocity of 1.052 m/s is twice the inlet and average velocity of 0.0526 m/s . b) The velocity profile from the CFD results is plotted at the inlet and at 45° (measured from the center of the turns) intervals, approximately 0.785 cm spacing, around 1.5 of the alternating 270° turns in the JIB. Showing the effect of the secondary flows on the velocity profile

of 100 ml/min (linear velocity is 0.0526 m/s and Re is 375.5). As shown in Figure 4a the center plane velocity has an oscillating pattern where the region of maximum velocity is continuously shifted toward the outside of the alternating 270° turns due to the generation of secondary flows. The velocity contours show that the location and magnitude of the maximum velocity at the centerline is constantly changing in response to the alternating curvature in the flow path deviating from the characteristic parabolic laminar velocity profile.

In Figure 4b, the magnitude of the velocity profile is plotted at 0.758 cm intervals of the 270° turns. At the inlet, where the flow path is straight, the velocity profile is the characteristic laminar flow parabolic profile where centerline velocity is 0.1052 m/s or twice the average of 0.0526 m/s , Figure 4b position 1. As previously shown in the velocity contours in Figures 4a and 4b graphically shows that the location of maximum velocity at the center plane shifts from left to right, moving toward the outer wall of the turns. The intervals plotted in Figure 4b further graphically show that the magnitude of the maximum velocity at the center

plane is reduced, when compared to the characteristic value 0.1052 m/s , and is as low as 0.076 m/s or 27.8% lower, as shown in position 3 in Figure 4. Therefore, despite operating in the laminar flow regime, $\text{Re } 375.5$, the alternating 270° turns with ROC of 1 cm successfully disrupted the characteristic parabolic laminar flow velocity profile and reduced the magnitude of the maximum centerline velocity.

For a more comprehensive evaluation of the JIB flow dynamics, the cross-sectional axial velocity contours and streamline patterns of the secondary flows were computationally evaluated at $\text{Re } 187.7$ and 375.5 (Figure 5). In Figure 5, images of the cross sectional velocity and secondary flow streamline patterns at turn centers and before and after flow inversion points show that the velocity profile and secondary flow pattern is highly unstable due to the flow path curvature and flow inversions. The secondary flow lines are symmetrical as the fluid reaches the center of the entrance turn and are then gradually more disordered, breaking up into uneven regions of swirling patterns, as the flow moves through

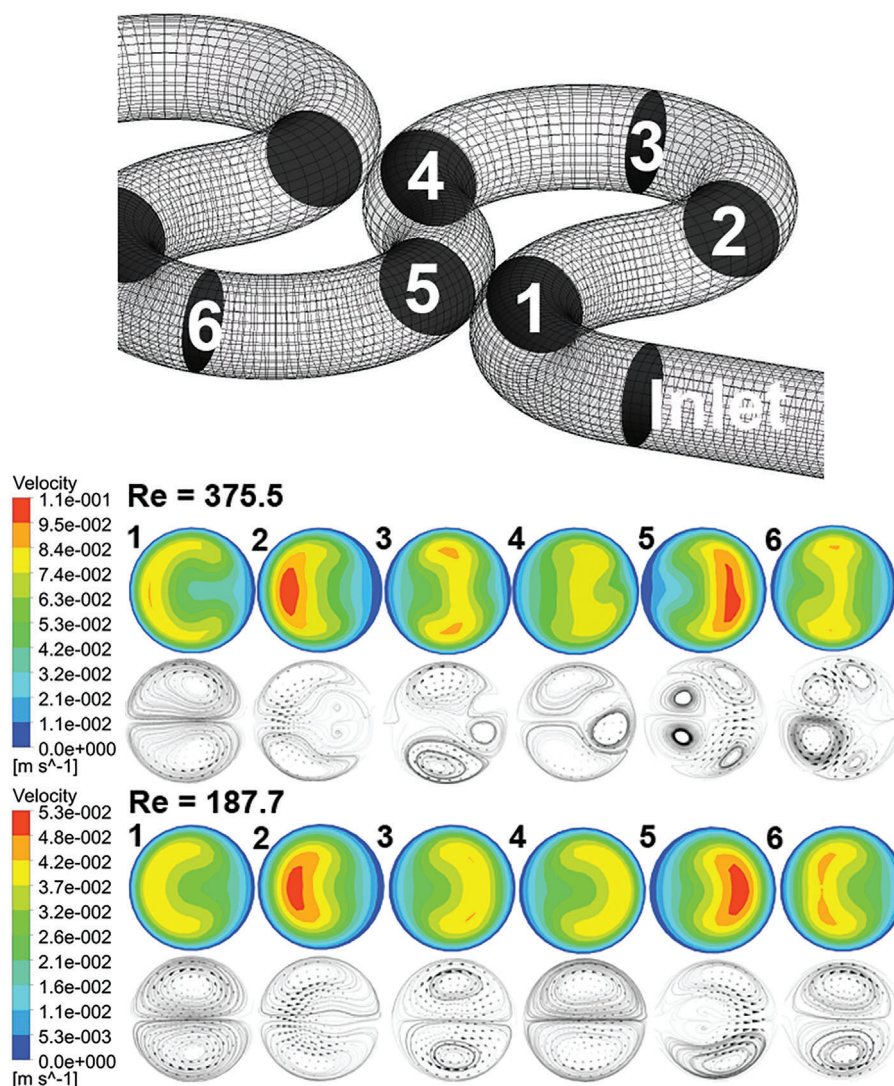


FIGURE 5 Computational fluid dynamics analysis of the cross-sectional velocity contours and secondary flow patterns for the Continuous Tubular Reactor (JIB) at two flow rates 100 ml/min (0.0526 m/s, Re 375.5) and 50 ml/min (0.0263 m/s, Re 187.7). The top image shows the measurement locations for the contours and streamlines, at the alternating 270° turn centers and at sections 90° from the turn centers. The bottom images are of the velocity contours and secondary flow streamlines at Re 375.5 and 187.7, showing the impact of centrifugal forces generated due to the curvature in the JIB flow path on the radial secondary flows and cross-sectional velocity

the reactor, this effect is more pronounced at higher Re, as seen in Figure 5. This non-symmetrical behavior of the secondary path lines is expected with larger values of Dean Number ($D > 100$) where the pressure differential is greater on the outside of the bend and thus the secondary flow has convected the axial momentum peak well away from the center of the cross-section (McConalogue & Srivastava, 1968).

Flow inversions change the direction of secondary flows and thus the location of the axial momentum peak, or maximum velocity, as seen in Figure 5 for both Re 187.7 and 375.5. The midpoint of the momentum peak shift from one side of the channel to the next occurs at the center of the alternating 270° turns, as shown in Figure 5 top and bottom sections 3 and 6. The cross-sectional

velocity contours in Figure 5 show that at this midpoint the region of maximum velocity is partitioned vertically, creating two peak velocity regions at the upper and lower halves of the channel, indicative of radial mixing.

As shown previously in the velocity profiles plotted in Figure 4 at position 3, as the flow approaches the turn inflection point (Figure 5 section 4) the maximum velocity decreases to approximately 0.041 and 0.079 m/s for Re 187.7 and 375.5, respectively. These values are 22% and 24.9% lower than characteristic maximum velocities for laminar flow at both Re values, respectively. Thus, the curvature of the flow path not only induces radial mixing, but also reduces the cross-sectional maximum velocity, both of these factors are critical to narrowing the RTD of the JIB.

5.2 | Residence time distribution

CFD tracer RTD simulations were conducted to predict the characteristic axial dispersion and minimum residence time (MRT) of the JIB before construction and were subsequently validated experimentally. These CFD simulations and experimental data were generated for a JIB length of 16.43 m containing 325 alternating 270° turns. As previously described, the MRT is the time when the first tracer fluid element exits the reactor, approximated as the time when the F curve, cumulative distribution curve, described in Equation (7) is equal to 0.005 ($F_{0.005}$) (Klutzn, Kurt et al., 2015; Orozco et al., 2017). The dimensionless E curves for the experimental and CFD Model were plotted for comparison, as shown in Figure 6, at 50 and 100 ml/min (Re 187.7 and 375.5). The empirical data was generated by averaging the results of three experimental runs for each flow rate.

To quantitatively determine the accuracy of the CFD model the variance (σ^2), dimensionless variance, dimensionless minimum residence time ($\theta_{0.5\%}$), dimensionless maximum residence time ($\theta_{99.5\%}$), relative width, mean residence time, and MRT values were computed and compared with the empirical values for two flow rates, 50 and 100 ml/min, as shown in Table 1. The variance values generated by the CFD model at 50 and 100 ml/min are within 11 and 15.8% of the experimental results, respectively, as shown in Table 1. The R_w of the CFD model at 50 and 100 ml/min was within 8.6 and 10.1% of the experimental data respectively. For 50 and 100 ml/min the CFD model predicted MRTs of 9.15 and 4.83 min which is within 4.9 and 3.6% of the experimental values of 9.62 and 5.01 min, respectively, Table 1. Thus, the CFD model performed with good accuracy, predicting the JIB MRT and RTD variance within 4.9 and 15.8% for our operating range of 50–100 ml/min.

5.3 | Pressure drop in JIB

In curved pipes the resistance to flow is greater than in straight pipes due to the outward centrifugal forces acting on the fluid causing a

radial pressure gradient. Thus the pressure drop in our JIB will be greater than that of a straight pipe of the same length. Our design objective here was to restrict the pressure drop value to a maximum of 5 psi. Pressure loss in the viral inactivation step would add to the overall pressure burden in the continuous downstream process, specifically impacting in-line chromatography steps which are highly sensitive to pressure. The pressure at varying JIB lengths from 16.43 to 148.5 m, corresponding to 1–9 JIBs (520 ml units), was measured experimentally with water at 50, 100, and 150 ml/min. The pressure values for 9 JIBs at 50, 100, and 150 ml/min (Reynolds numbers 187.7, 375.5, and 563.2) were 0.74, 1.9, and 3.44 psi, respectively. This is 64%, 110%, and 154% higher than the theoretical values of 0.451, 0.902, 1.353 psi for a straight tube under the same conditions, calculated via the Hagen–Poiseuille law. Demonstrating that while the curvature increases mixing in the system, it also contributes to a substantial increase in the pressure drop and this effect is more pronounced at higher flow rates. However, within our operating range of 50–100 ml/min the pressure drop is below 5 psi for lengths that provide 60 min incubation times, which corresponds to approximately 66 m and 0.45 psi to 214.5 m and 2.8 psi, respectively.

5.4 | Scaling

To determine how the JIB RTD and MRT scales with length, surface monitors were placed at intervals along the JIB flow path of the CFD model to measure the simulated RTDs at 50 ml/min. As shown in Figure 7a the CFD model predicted a linear relationship of length versus variance with a slope of 0.0452. For experimental validation of the CFD model, pulse tracer experiments were conducted for JIBs connected in series, as shown in Figure 3d, from 1 to 7 units, or 16.43 to 115 m, which consist of 325–2275 (270°) turns, at 50 ml/min or Re 187.7. As shown in Figure 7 the experimental results demonstrated a linear relationship between length and

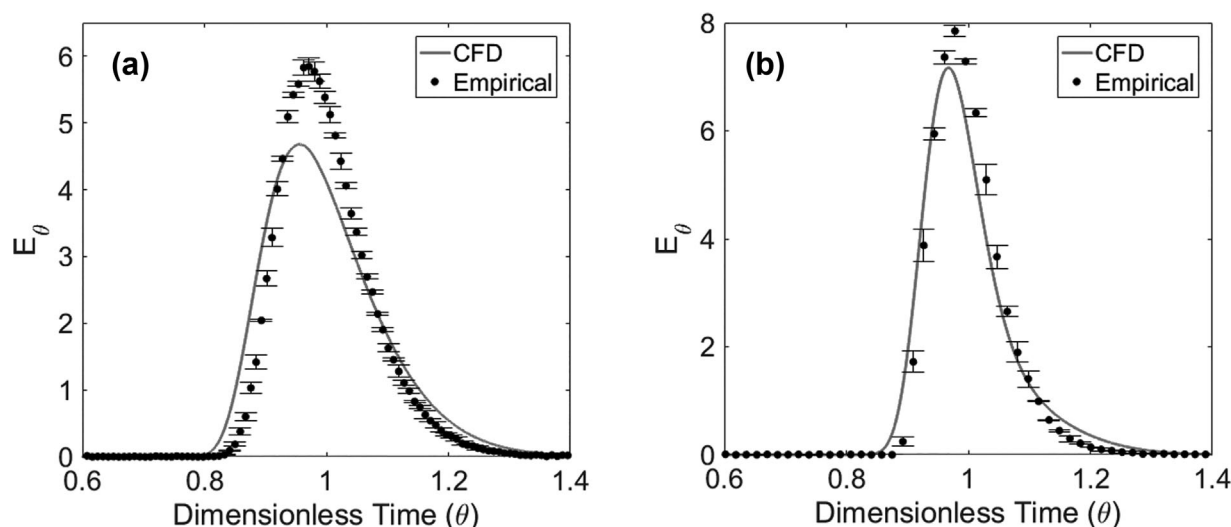


FIGURE 6 Comparison of CFD Model and tracer study dimensionless $E(\theta)$ Residence Time Distribution (RTD) curves for Continuous Viral Inactivation Reactor (JIB) for (a) 50 ml/min (0.0263 m/s and Re 187.7) and (b) 100 ml/min (0.0526 m/s and Re 375.5) flow rates

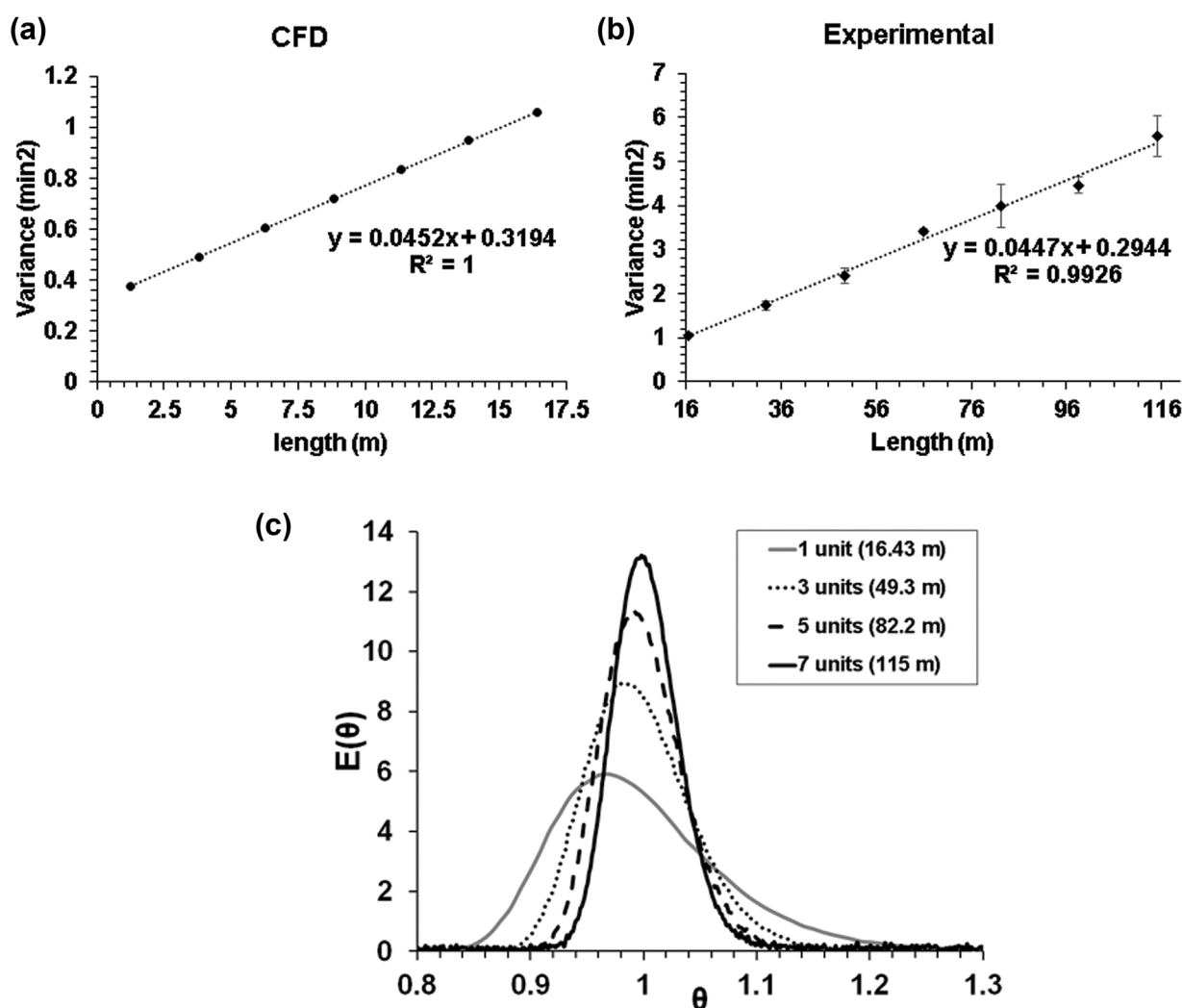
TABLE 1 Experimental data and CFD Model Comparison of the JIB RTD characteristic values of the dimensionless variance (σ_θ^2), dimensionless minimum residence time ($\theta_{0.5\%}$), dimensionless max residence time ($\theta_{99.5\%}$), relative width, and minimum residence time at 50 and 100 ml/min

Flow rate	Source	Variance, σ^2 (min ²)	Variance, σ_θ^2	$\theta_{0.5\%}$	$\theta_{99.5\%}$	Rw	t_{mean} (min)	MRT, t_{min} (min)
50 ml/min (0.0263 m/s)	CFD	1.09	8.96E-03	0.831	1.317	0.631	11.01	9.15
	Experimental	1.06 ± 0.1	8.32E-03	0.877	1.280	0.685	11.27 ± 0.01	9.4 ± 0.1
100 ml/min (0.0526 m/s)	CFD	0.16	5.39E-03	0.877	1.280	0.685	5.51	4.83
	Experimental	0.19 ± 0.01	3.18E-03	0.876	1.149	0.762	5.72 ± 0.01	5.01 ± 0.3

variance with a slope of 0.0447, shown in Figure 7b. The CFD model slope was within 1.12% of experimental value. This shows the CFD model can be used to accurately predict the RTD and MRT of the JIB at various lengths.

Additionally, the dimensionless RTD curves, plotted in Figure 7c, show the profile becomes more symmetrical and tighter about the

mean residence time ($\theta = 1$) as the reactor length increases, indicating a smaller deviation from plug flow with increasing JIB length. This convergence toward plug flow with increasing length is possibly due to tracer elements sampling more radial positions, and therefore more velocity profiles, causing each tracer element to approach the same mean velocity. This means that the extra volume needed beyond the

**FIGURE 7** Scaling JIB with Length at 50 ml/min (0.0263 m/s and $Re = 187.7$). CFD and Experimental pulse tracer studies of the Residence Time Distribution variance (σ^2), the spreading of pulse as it passes through the JIB versus length were used to determine variance at varying JIB lengths at 50 ml/min. a) CFD variance vs JIB length from 0 to 16.43 m. b) Experimental variance versus length, for JIB lengths from 16.43 to 115 m. c) The dimensionless RTD Curves for 1 (16.43 m), 3 (49.3 m), 5 (82.2 m), and 7 (115 m) JIB lengths, from the experimental pulse tracer studies. The x-axis represents dimensionless time (θ) and the y axis normalized exit age function $E(\theta)$

ideal plug flow volume (where $t_{\min} = t_{\text{mean}}$) decreases with increasing length. Thus, the smallest volume plug flow reactor would be one that has the maximum length allowed given the pressure limitations imposed. One can envision optimizing tube diameter, diameter of curvature, and length for different flow rates to achieve the maximum allowable pressure, to achieve the plug flow reactor with the smallest volume. Even so, the current design is close enough to optimal for the flow rates tested. The $t_{\min}$ is only 15% smaller than the t_{mean} (see Table 1). Thus, further reduction in JIB volume is not warranted. Additionally, the MRT for 7 units or JIB reactor length of 115 m, determined from experimental data as shown in dimensionless form in Figure 7, was 70.4 ± 0.46 min, only 2.44 min from the plug flow value of 72.84 min for the same volume (Experimental volume 3,642 ml) at the same 50 ml/min flow rate, where plug flow is the V/Q , where V is volume and Q is flow rate.

6 | CONCLUSION

A continuous flow tubular reactor, the JIB, for continuous low pH viral inactivation, was designed, evaluated using computational fluid dynamics, and 3D printed for experimental validation. The JIB design requirements were generated with the goal of limiting the reactor pressure drop and streamline integration into the downstream purification process: ii) a ≤ 5 psi pressure drop along the length of the tube, (ii) achieve radial mixing to minimize axial dispersion without moving parts, and (iii) operate at the laminar flow regime. Additionally, a minimum residence time (MRT) approach was used to ensure that the desired product holding time was met.

The flow path of the JIB was designed in a serpentine pattern with alternating 270° turns. The CFD model was used to study and quantify the effect of the dean vortices generated by the curved JIB flow path geometry on the RTD. CFD model results showed that the curvature in the flow path resulted in the generation of dean vortices and confirmed that changes in the flow direction disrupted the velocity profile. These secondary flows and momentum shifts in the velocity profile are indicative of passive mixing and consequently lead to reduced axial dispersion and narrow the RTD when compared to laminar flow in a straight pipe of the same length. The CFD model accuracy was demonstrated quantitatively comparing the experimental and computational values for the variance (σ^2), dimensionless MRT ($\theta_{0.5\%}$), and dimensionless maximum residence time ($\theta_{99.5\%}$). The variance values were within 15.8% accuracy between the model and the experimental data.

The CVI reactor, JIB, was characterized in terms of flow dynamics, pressure drop, and RTD at a range of flow rates. The design is compact, flexible, modular, and scalable, which allows for easy process integration. The RTD curves show a close to plug flow behavior and the MRT is easily determined via the F curve response and can be predicted using a CFD model.

ORCID

Stephanie A. Parker  <http://orcid.org/0000-0001-9822-8887>

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